

Technology Stack

Date	16 September 2026
Team ID	SWTID-2026-2311
Project Name	AI StudyBuddy API
Maximum Marks	2 Marks

Technology Stack Details

AI StudyBuddy is a headless RESTful backend; a React.js client (in progress alongside this cycle) and API tools (Postman / Thunder Client) both consume the same REST endpoints during development and grading.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js client (register/login, upload material, view AI-generated resources) plus Postman/Thunder Client for direct API verification	React gives students an interactive dashboard, while Postman/Thunder Client let the team verify every endpoint independently of frontend progress
2	Backend / Server-Side	Node.js + Express.js, with <code>cors</code> (origin from <code>CLIENT_URL</code> or <code>http://localhost:3000</code>) and <code>express-async-errors</code>	Non-blocking I/O suits the Gemini AI calls; <code>express-async-errors</code> lets async controller errors flow straight into the centralized error handler
3	Database / Data Storage	MongoDB Atlas (managed cloud cluster) with Mongoose ODM	A managed Atlas cluster removes local database administration for the team; Mongoose adds schema validation and structured querying on top of raw MongoDB documents
4	Cloud / Hosting / Deployment	API server run on port 5000 (<code>process.env.PORT 5000</code>) against the team's MongoDB Atlas cluster (<code>MONGO_URI</code>) for the 15–29 September 2026 grading window; fully <code>.env</code> -driven configuration	Keeps credentials out of source control while already running against a production-grade managed database, simplifying a future full cloud deployment
5	Version Control & CI/CD	Git & GitHub – github.com/sachinselva038-nm/studybuddy_selva	Single source of truth for the 4-member team; commit history traces each Epic/Sprint
6	Third-Party APIs / Other Tools	Google Gemini AI (<code>@google/generative-ai</code> , <code>model gemini-2.5-flash</code>) for summary/flashcard/quiz/study-plan generation; <code>jsonwebtoken</code> + <code>bcryptjs</code> (auth/security); <code>multer</code> (file upload); <code>dotenv</code> , <code>cors</code> , <code>nodemon</code> (dev)	Each library enables one specific, named feature of the solution — generative learning content, secure auth, safe file intake, config isolation, and cross-origin support